WEEKLY INTERNSHIP PROGRESS REPORT

Company: Codexus Technologies
Project: XIVIX – Scalable Humanoid Robot Platform
Position: Electronics and IoT Trainee
Reporting Period: 03 November – 06 November 2026
Week: 01
Project Phase: Research, Requirements & Concept Development

1. Weekly Objectives

- Understand the XIVIX humanoid robot project's objectives, technical requirements and development roadmap.
- Set up the necessary engineering software and development environment.
- Study fundamental humanoid robotics concepts, mechanical structures and actuator technologies.
- Explore existing humanoid robot designs and available CAD models.

2. Activities Completed

Project Introduction and Planning

- Discussed the six-month internship plan, project scope and expected deliverables with the project supervisor.
- Reviewed the XIVIX Robot Project Brief to understand the proposed system architecture, scalability requirements and development phases.
- Identified the major engineering areas involved, including mechanical design, embedded systems, electronics, motion control and IoT.

Development Environment Setup

- Set up FreeCAD for mechanical design and 3D modelling.
- Set up KiCad for electronic schematic and PCB design.
- Configured PlatformIO IDE for embedded firmware development.
- Set up Webots for robotics simulation.

Humanoid Robotics Research

- Studied the basic structure and degrees of freedom (DOF) of humanoid robots.
- Reviewed servo motors and industrial actuators motors for robotic joint actuation.
- Explored fundamental concepts related to robot balance, centre of mass and joint movement.
- Studied communication protocols including UART, I²C, SPI, CAN and MQTT.

Mechanical Design Exploration

- Explored the open-source Poppy Humanoid robot project.
- Obtained and examined available CAD models of humanoid robot components.
- Imported and inspected robot CAD files using FreeCAD to understand mechanical structures and assemblies.

3. Knowledge and Skills Gained

- Improved understanding of humanoid robot architecture and mechanical joint configurations.
- Gained initial experience using FreeCAD for inspecting robotic mechanical designs.
- Developed familiarity with robotics simulation and embedded development tools.
- Strengthened theoretical knowledge of actuators and communication protocols used in robotic systems.

4. Challenges Encountered

- Understanding the complexity of humanoid mechanical structures and joint arrangements.
- Becoming familiar with FreeCAD's modelling workflow and imported CAD files.
- Identifying suitable actuator technologies for a compact humanoid robot.

5. Plan for the Following Week

- Study existing humanoid robot platforms and compare their mechanical designs.
- Continue practising FreeCAD through robotic component modelling.
- Investigate servo motor dimensions, mounting mechanisms and joint brackets.
- Explore robot stability, weight distribution and actuator torque requirements.
- Begin preliminary mechanical design experiments.

6. Weekly Summary

During the first week, the main focus was on project familiarisation, development environment preparation and fundamental robotics research. The activities established an initial understanding of the XIVIX humanoid robot's technical requirements and provided a foundation for subsequent mechanical design, actuator selection and simulation work.